2. Step-by-Step Creation Process

Step 1: Setting Up the Pool and Swimlanes

I opened draw.io, enabled the BPMN shape library, and created a horizontal pool labeled *SYSTEM ILLUSTRATION : SUBSIDIZED INPUT DISTRIBUTION*. Inside the pool, I added four swimlanes to clearly separate human responsibilities from automated database and reporting layers: (1) Small Holder Farmer, (2) Cooperative Store Manager, (3) Oracle Database, and (4) Analytics Dashboard.

Step 2: Defining the Start Event and Initial Verification

In the Small Holder Farmer lane, I placed a green Start Event circle (*Farmer requests Subsidized Inputs*) and connected it with a sequence flow arrow to the first task box (*ACTIVITY MANAGER Verifies Member ID & INPUT Request*). This establishes how a request enters the cooperative system.

Step 3: Adding Data Retrieval and Decision Logic

Next, I routed the flow down into the Cooperative Store Manager lane. I added a cylindrical Data Store symbol (*Data Store Retrieve*) linked to an activity task (*Retrieve system Queries Farmer Quota & Stock levels*). To handle approval rules, I inserted an Exclusive Gateway diamond (*Gateway Quota & Stock Valid*) with two outgoing paths: **No** routes to *Activity Reject / Adjust Request & Notify Farmer*, while **Yes** proceeds directly to database transaction processing.

Step 4: Modeling Oracle Database Operations and PL/SQL Triggers

In the Oracle Database lane, I modeled three core CRUD activities: `[CREATE] Activity Update Distributions Table`, `[UPDATE] Activity Decrement Stock Quantity in Inputs Table`, and `[CREATE] Activity Log Transaction in Ledger`. Using dotted association lines and annotation blocks, I attached three PL/SQL trigger definitions (`AFTER INSERT` and `AFTER UPDATE` on stock and transaction tables) to show how inventory counts and audit ledger entries are updated automatically.

Step 5: Connecting Real-Time Analytics and the End Event

Finally, in the Analytics Dashboard lane, I linked the database operations via *Data Feed* and *Real-Time Update* connectors to two reporting tasks: *Activity Refresh Regional Distribution Metrics* and *Activity Update Stock level Alerts*. The workflow concludes at a red End Event circle (*Transaction complete & Dashboard Updated*).

3. Summary of Swimlanes & BPMN Elements

Swimlane / Layer	BPMN Elements Used	Role in the Process
1. Small Holder Farmer	Start Event (Green Circle), Task Box	Initiates the subsidized input request and undergoes Member ID and request verification.
2. Cooperative Store Manager	Data Store, Task Boxes, Exclusive Gateway (Diamond)	Queries farmer quota and stock availability, validates eligibility, and approves, adjusts, or rejects requests.
3. Oracle Database	CRUD Task Boxes, Dotted Associations, Trigger Annotations	Inserts distribution records, decrements stock quantities, logs ledger entries, and fires PL/SQL `AFTER INSERT` / `AFTER UPDATE` triggers.
4. Analytics Dashboard	Task Boxes, Data Feed Flows, End Event (Red Circle)	Refreshes regional distribution metrics, triggers low-stock alerts in real time, and marks transaction completion.